Exception Handling in Java

Information regarding Exception:-

- Dictionary meaning of the exception is abnormal termination.
- An unexpected event that disturbs or terminates normal flow of execution called exception.
- If the application contains exception then the program terminated abnormally the rest of the application is not executed.
- To overcome above limitation in order to execute the rest of the application must handle the exception.

Exception Handling in Java

In java we are having two approaches to handle the exceptions.

1) By using try-catch block.
2) By using throws keyword.

Exception Handling:-

The main objective of exception handling is to get normal termination of the application in order to execute rest of the application code.

Exception handling means just we are providing alternate code to continue the execution of remaining code and to get normal termination of the application.

Exception Handling in Java

Types of Exceptions:-

As per the sun micro systems standards The Exceptions are divided into three types

1) Checked Exception
2) Unchecked Exception
3) Error

Exception Handling in Java

checked Exception:-

- ✓ The Exceptions which are checked by the compiler at the time of compilation is called Checked Exceptions.
  IOException,SQLException,InterruptedException……..etc
- ✓ If the application contains checked exception the code is not compiled so must handle the checked Exception in two ways
  - o By using try-catch block.
  - o By using throws keyword.
- ✓ If the application contains checked Exception the compiler is able to check it and it will give intimation to developer regarding Exception in the form of compilation error.

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Unchecked Exception:-

- The exceptions which are not checked by the compiler at the time of compilation are called unchecked Exception. ArithmeticException,ArrayIndexOutOfBoundsException, NumberFormatException ……etc.
- If the application contains un-checked Exception code is compiled but at runtime JVM(Default Exception handler) display exception message then program terminated abnormally.
- To overcome runtime problem must handle the exception in two ways.
  - By using try-catch blocks.
  - By using throws keyword.

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Error:-

- Errors are caused due to lack of system resources like,

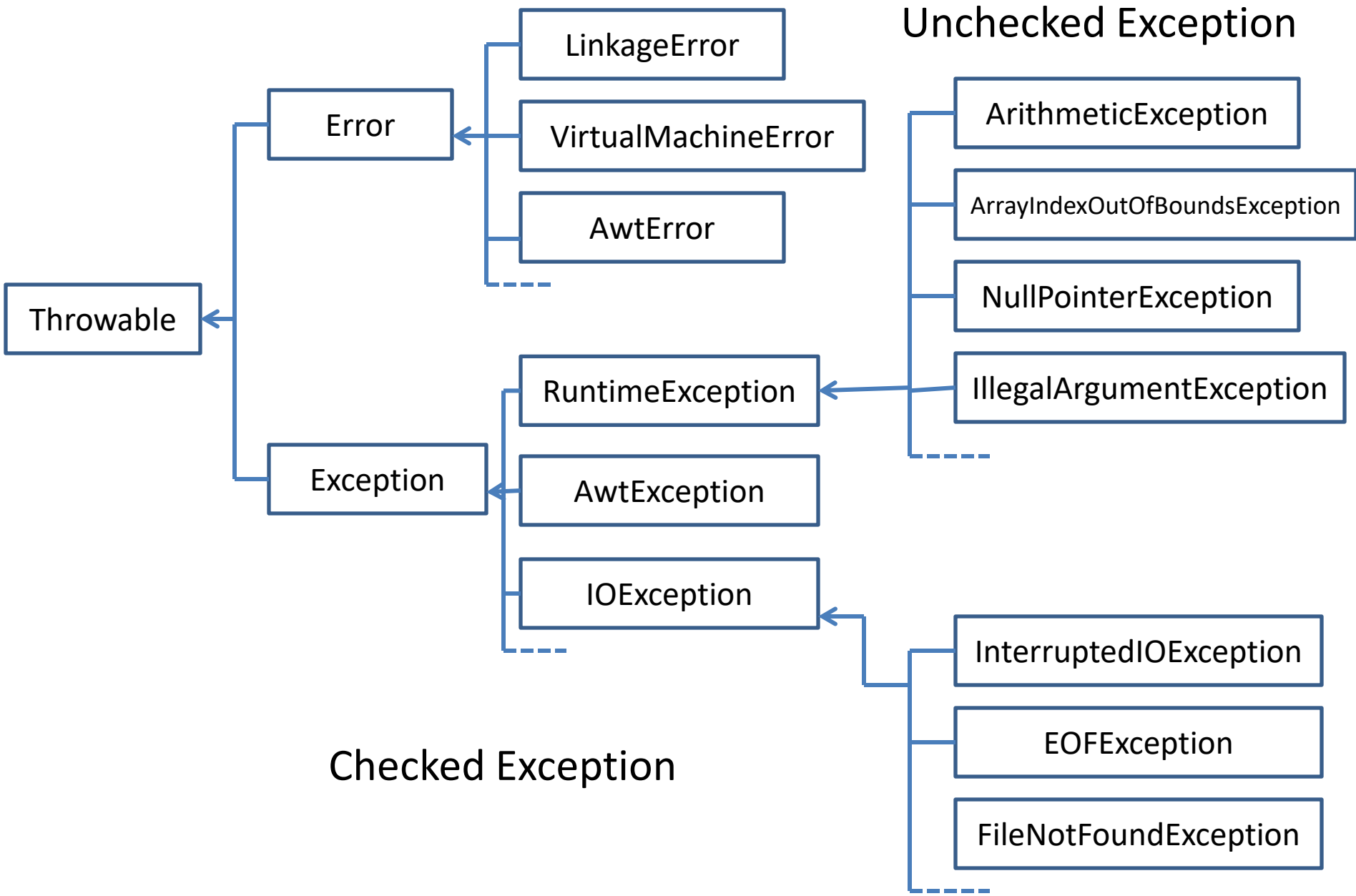
o Heap memory full.

o Stack memory problem.

o AWT component problems…..etc

Ex: - StackOverFlowError, OutOfMemoryError, AssertionError…etc

- Exceptions are caused due to developers mistakes or end user supplied inputs but errors are caused due to lack of system resources.
- We are handle the exceptions by using try-catch blocks or throws keyword but we are unable to handle the errors


Unchecked Exception
LinkageError
Error
VirtualMachineError
AwtError
ArithmeticException
ArrayIndexOutOfBoundsException
NullPointerException
Throwable
RuntimeException
IllegalArgumentException
Exception
AwtException
IOException
InterruptedIOException
EOFException
FileNotFoundException
Checked Exception

Exception handling key words:-

1)try

2) catch

3) finally

4) throw

5) throws

Exception handling by using Try –catch block:- Syntax:-

```
try
{
exceptional code;
}
catch (ExceptionName reference_variable)
{
Code to run if an exception is raised;
}
```